COL7378: Principles of Autonomous Systems
Semester II, 2025-26

Course Introduction

Rohan Paul

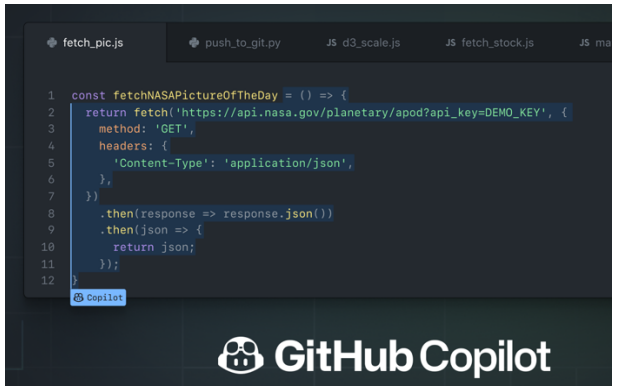
Today’s lecture

- Last Class
  - Course Organization
- This Class
  - Introduction
    - What is embodied AI?
    - Core challenges: Estimation and Planning

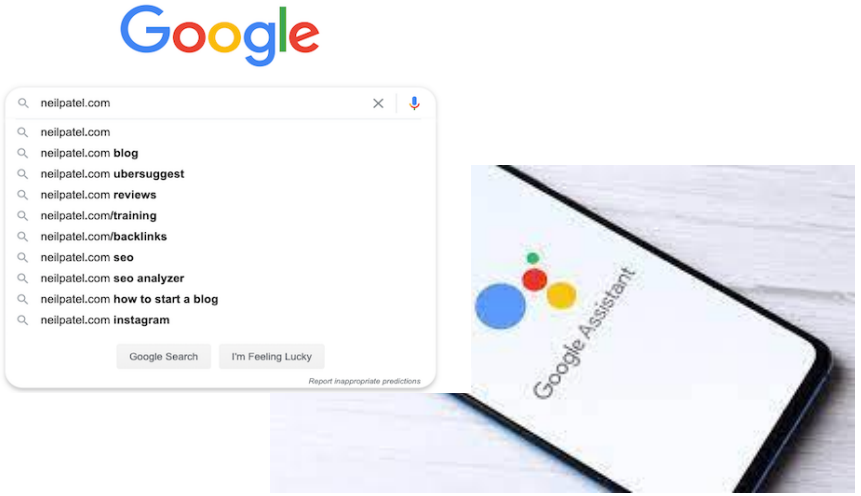
Successful AI Deployment

Audio to Concepts

Description to code



Partial text – full text



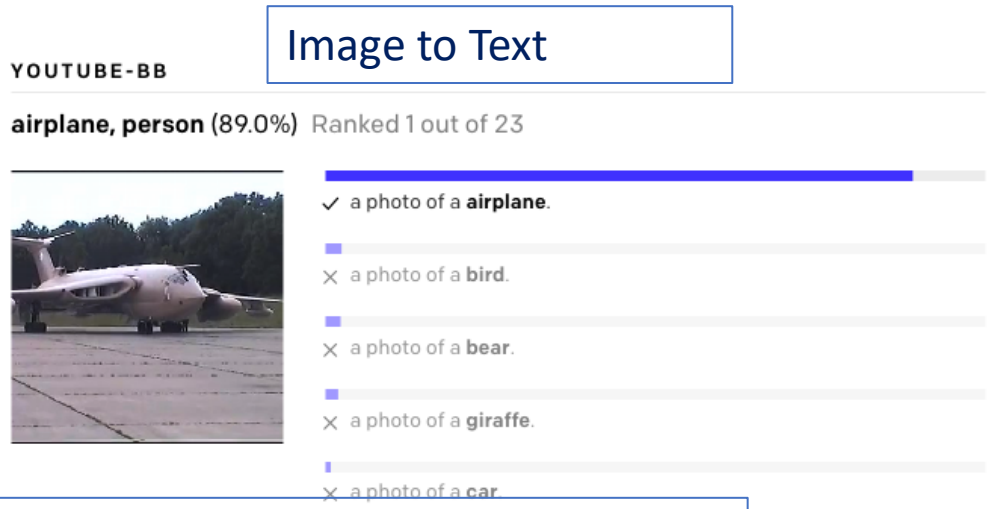
Text to Audio

Deep Speech: Scaling up end-to-end speech recognition

Awni Hannun, Carl Case, Jared Casper, Bryan Catanzaro, Greg Diamos, Erich Elsen, Ryan Prenger, Sanjeev Satheesh, Shubho Sengupta, Adam Coates, Andrew Y. Ng

Baidu Research – Silicon Valley AI Lab

Image to Text



Successes in learning functions from raw sensing to representations that carry semantics

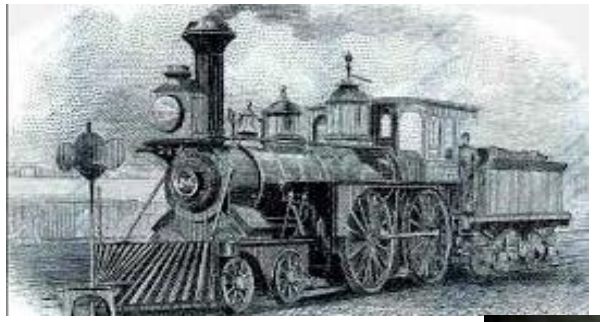
The Evolution of Machines in Society

Agricultural Age

Plough

Industrial Age

Engine



Electronics Age

Computers

Robotic age

Sensing & compute

Increased capacity to sense interact and manipulate the physical world

Emergence of Embodied AI

Modality specific representations

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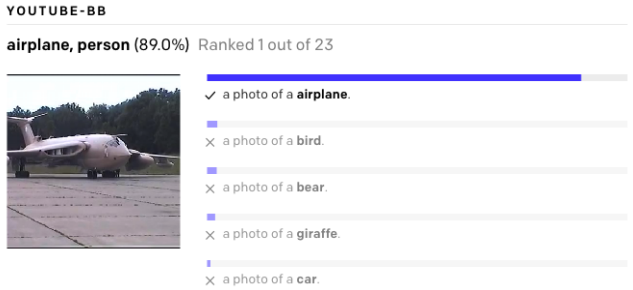


Embodied AI

Intelligent agents capable of autonomous goal-directed behaviour

World Modeling, Reasoning and Planning, Imitation.

Physical interaction/ exchange of energy

Embodied AI Systems: An Emerging Future

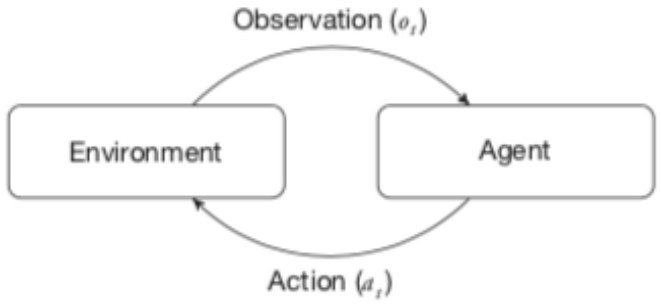
Caring for the infirm

Factories of the future

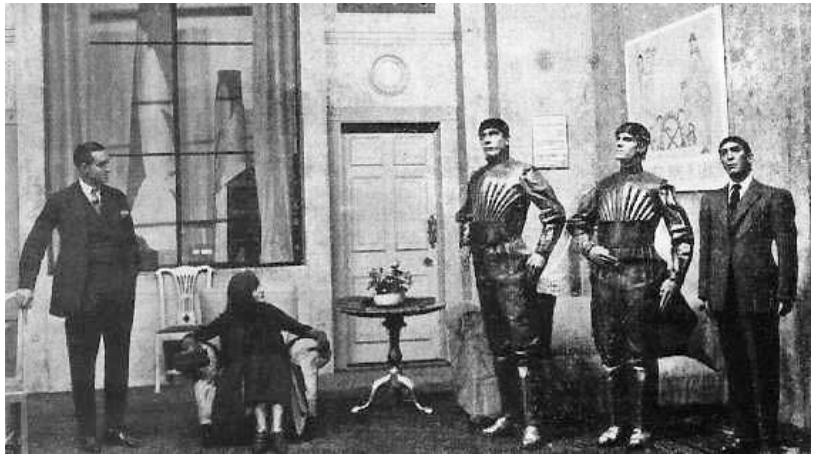
Security and first responders

Historical Context

- Intelligent agents that can perform tasks in the physical world.
- Platforms that can carry a payload, sense the environment and take actions.
- Perform goal-directed tasks in the real world.
- Human interaction – task specification, resolving ambiguity etc.
- Various names: Embodied AI, autonomous systems, robot intelligence.
- The word “Robot” means “labour” in Czech by Karl and Joseph Kapek (1920s).



At a basic level, an embodied AI agent takes observations from the environment and synthesizes goal-directed actions.



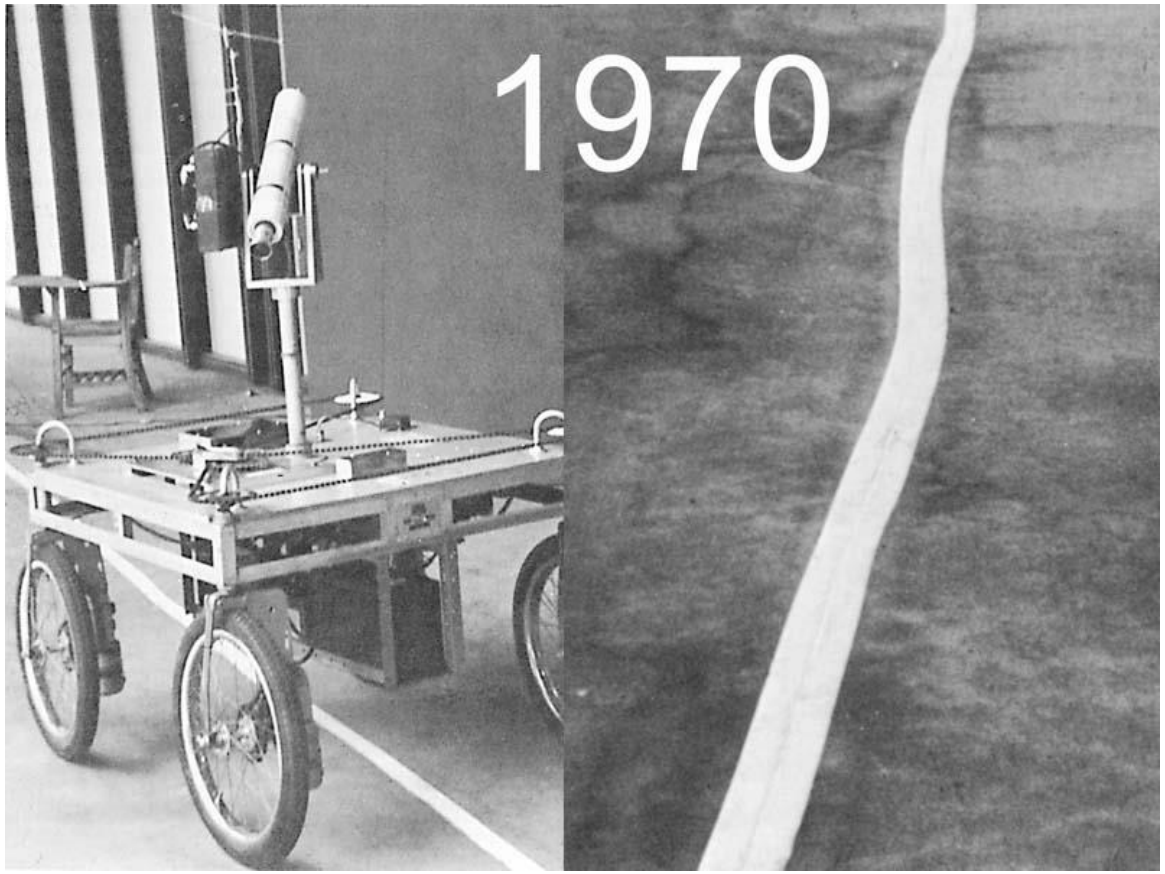
A scene from “Rossum’s Universal Robots,” showing three robots, https://www.sciencefriday.com/segments/the-origin-of-the-word-robot/

Historical Perspective

- Quest to create artificial agents that can sense, reason and act intelligently in the world.
- Stanford's Cart (Hans Moravec) following a line.

Obstacle avoidance and navigation in the real world by a seeing robot rover

Responsibility	by Hans Peter Moravec.
Imprint	1980.
Physical description	vii, 170 leaves, bound : ill. ; 28 cm.

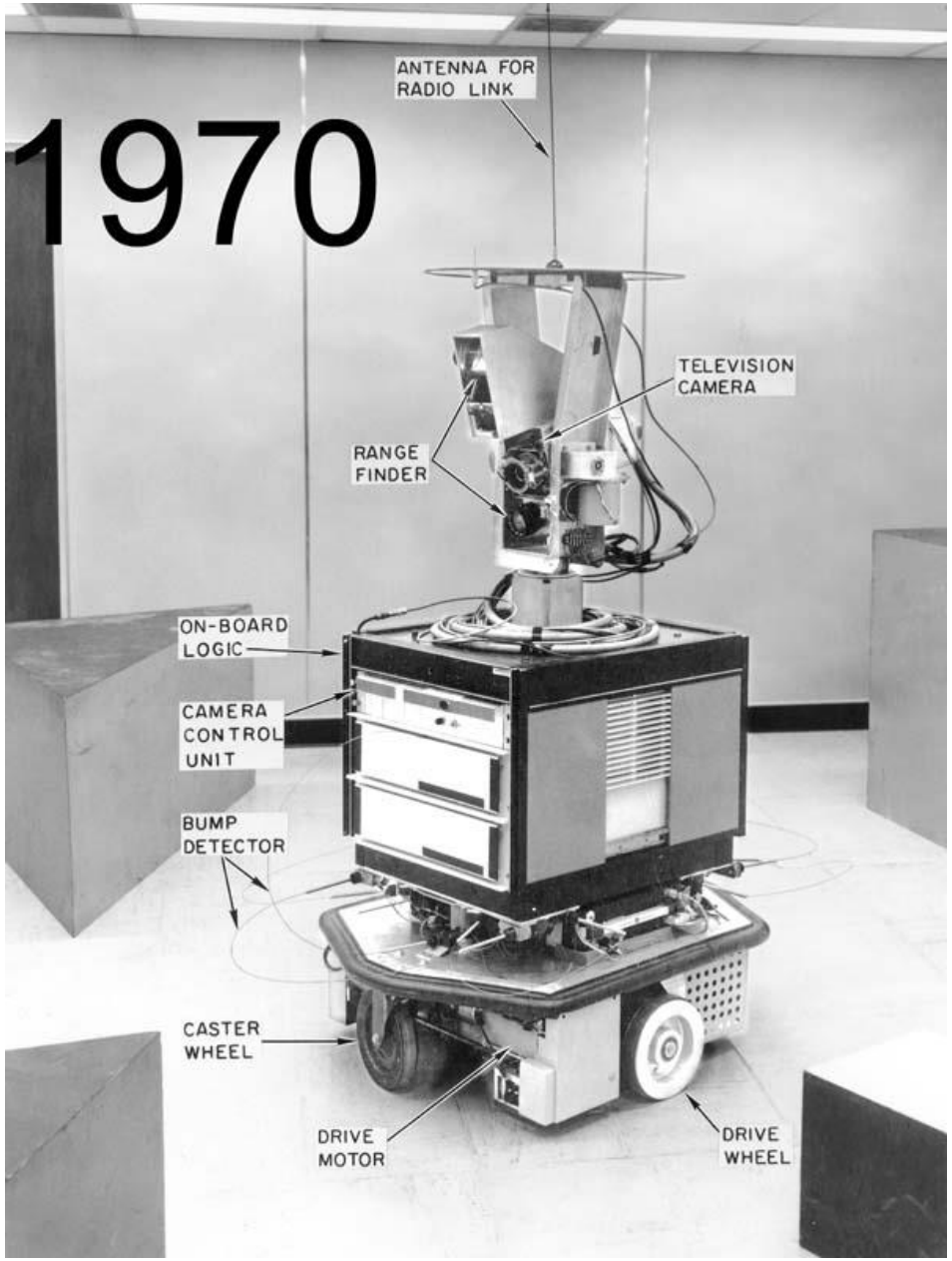


https://youtu.be/8Mxk2L3lu9Q

[redacted]

Historical Perspective

- Lab Experiments
  - Create artificial agents that take high-level goals and move in the environment
- Shakey
  - Stanford Research Institute (SRI)
  - The A* planner (Nils Nilsson) emerged from this research project



https://youtu.be/GmU7SimFkpU

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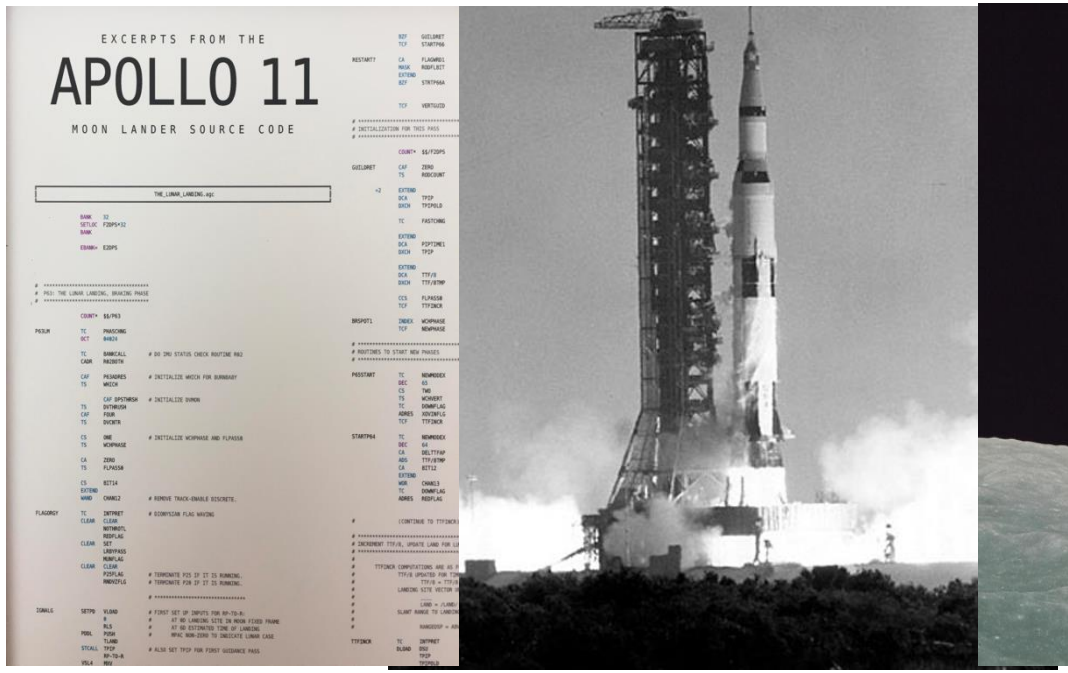
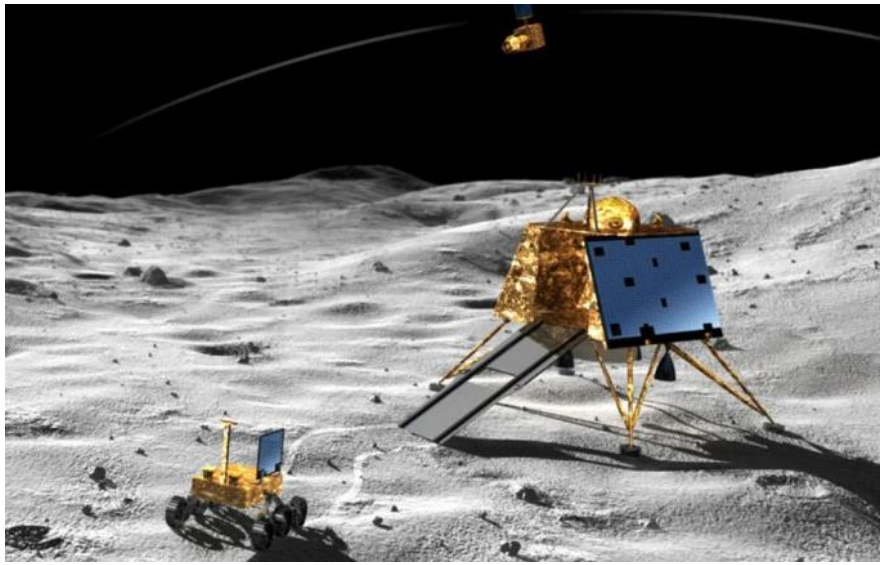
Precision Guidance and Navigation

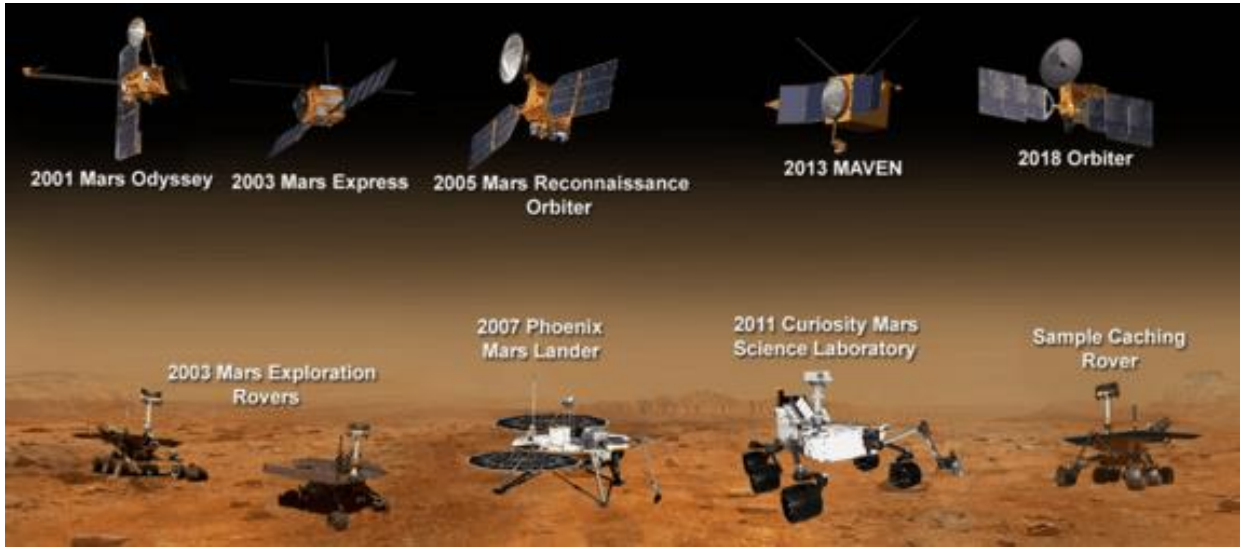
- Computer Guidance
  - 1960s.
  - Computers controlling systems
- Task
  - Fuse all the sensor data,
  - Determine the system state and take actions
- Demonstration
  - Apollo program
  - Computers performed that allow precise state estimation and guided navigation.
- Key Innovation
  - Kalman Filtering (later in the course)
  - Advances in handling uncertainty in measurements and actions.



Other Examples

- Complex operations in remote and hazardous environments.





https://www.youtube.com/watch?v=N9hXqzkH7YA

Modern Examples: Autonomous Driving

- Vehicles capable of detecting roads, people, traffic signs, following lanes, avoiding collisions etc.

Darpa Grand Challenge (2005)
https://www.youtube.com/watch?v=M2AcMnfzpNg

Industry Automation

- Manufacturing Section
  - Precision
  - Speed
  - Sectors: semi-conductor, heavy engineering (engines), automotive.
- Factory robots
  - Computer controlled devices.
  - Pre-programmed trajectories.
  - Limitation
    - Very limited reasoning and hence cannot adapt easily.

Robot: PickUp(1.5), Weld(2s), Put(2.0), … Repeat

Example: Orangewood Robotics
https://twitter.com/i/status/1685310025802432518

Ware house Automation

Amazon Robotics

Addverb Automation

Grey Orange Robotics

Need for reasoning

Vehicle on the road that can detect and avoid obstacles.

Vehicle that can understand that a ball coming on the road may be followed by a child.

Factory robot that can pick and place tools.

Factory robot that can be instructed to assemble a solar panel

A robot in the hospital that can deliver samples.

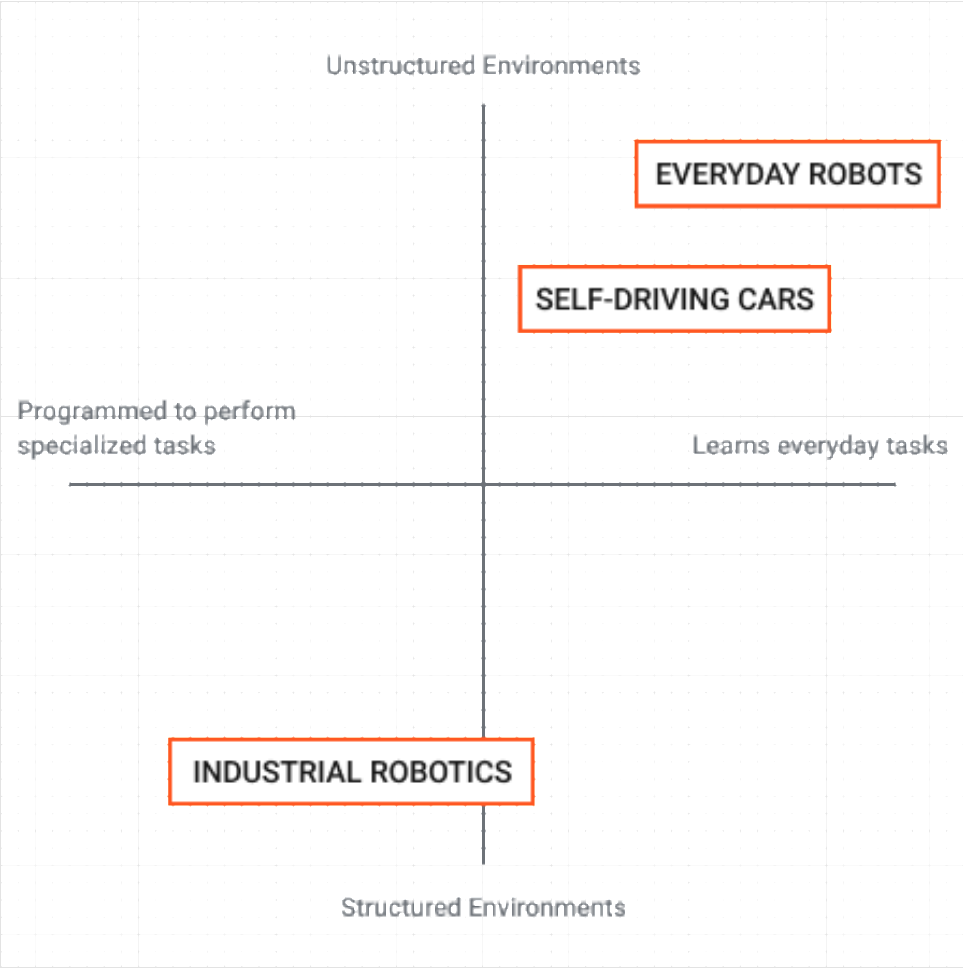
A robot that can assist the surgeon by watching the surgery and providing tools, responding to contingencies.

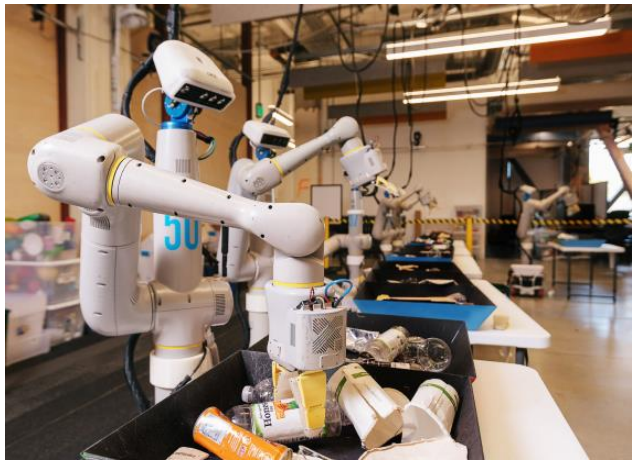
A remote operated vehicle that can relay images.

A vehicle that can be commanded to survey behind the wall, inform if there is a person, deliver supplies.

Writing explicit rules is going to be challenging..... *Uncertainty*
Reason about long-term consequence of actions ..... *Planning*

Performing General Purpose Tasks



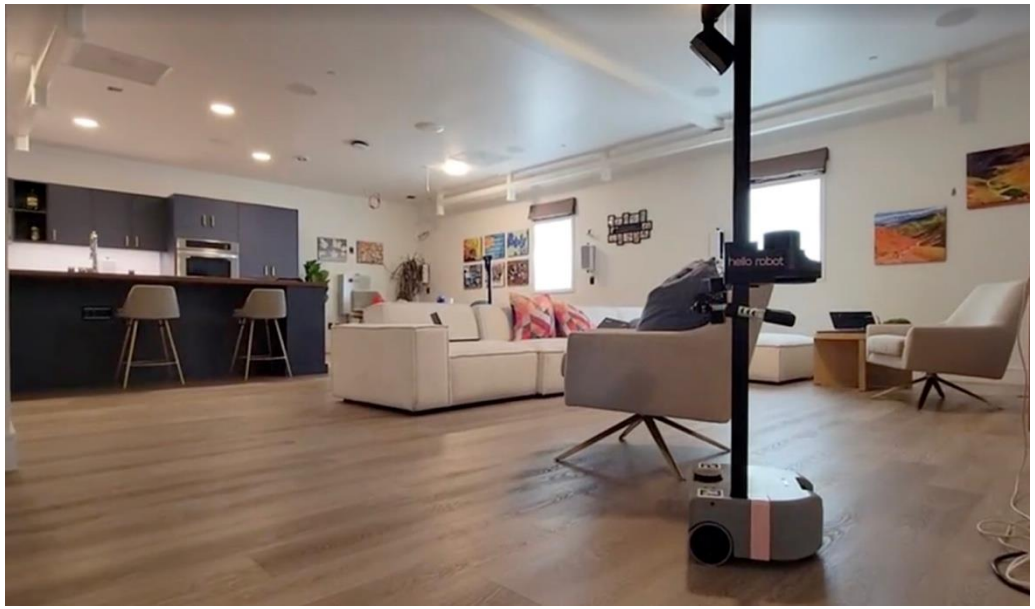


https://x.company/projects/everyday-robots

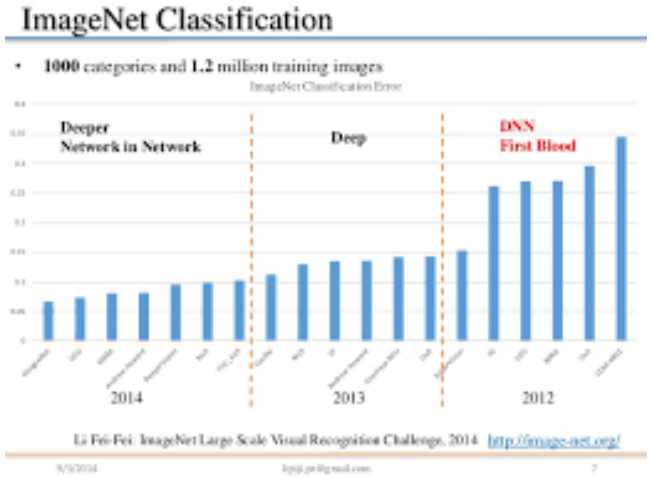
Example

Google Robotics Research
https://sites.research.google/palm-saycan

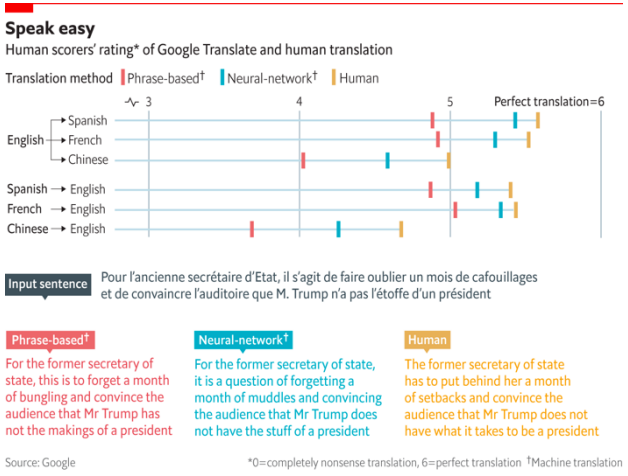
Facebook AI Research
https://aihabitat.org/challenge/2023_homerobot_ovmm/

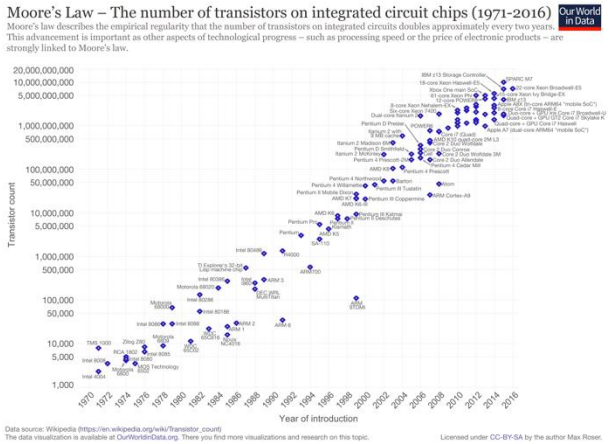


General Purpose Intelligence



Embodied AI systems confront many problems together.
Require general-purpose intelligence.





Intelligence measures an agent's ability to achieve goals in a wide range of environments

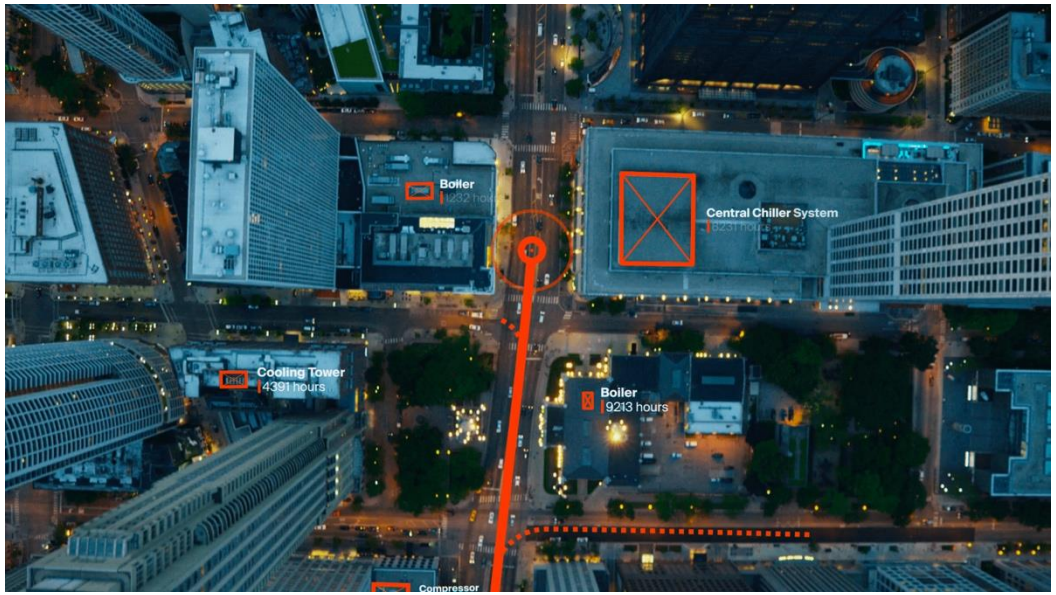
$$\Upsilon(\pi) := \sum_{\mu \in E} 2^{-K(\mu)} V_\mu^\pi .$$

Measure of Intelligence — Sum over environments — Complexity penalty — Value achieved

Universal Intelligence: A Definition of Machine Intelligence, Legg & Hutter

Other Embodied AI Systems

Voice interaction systems

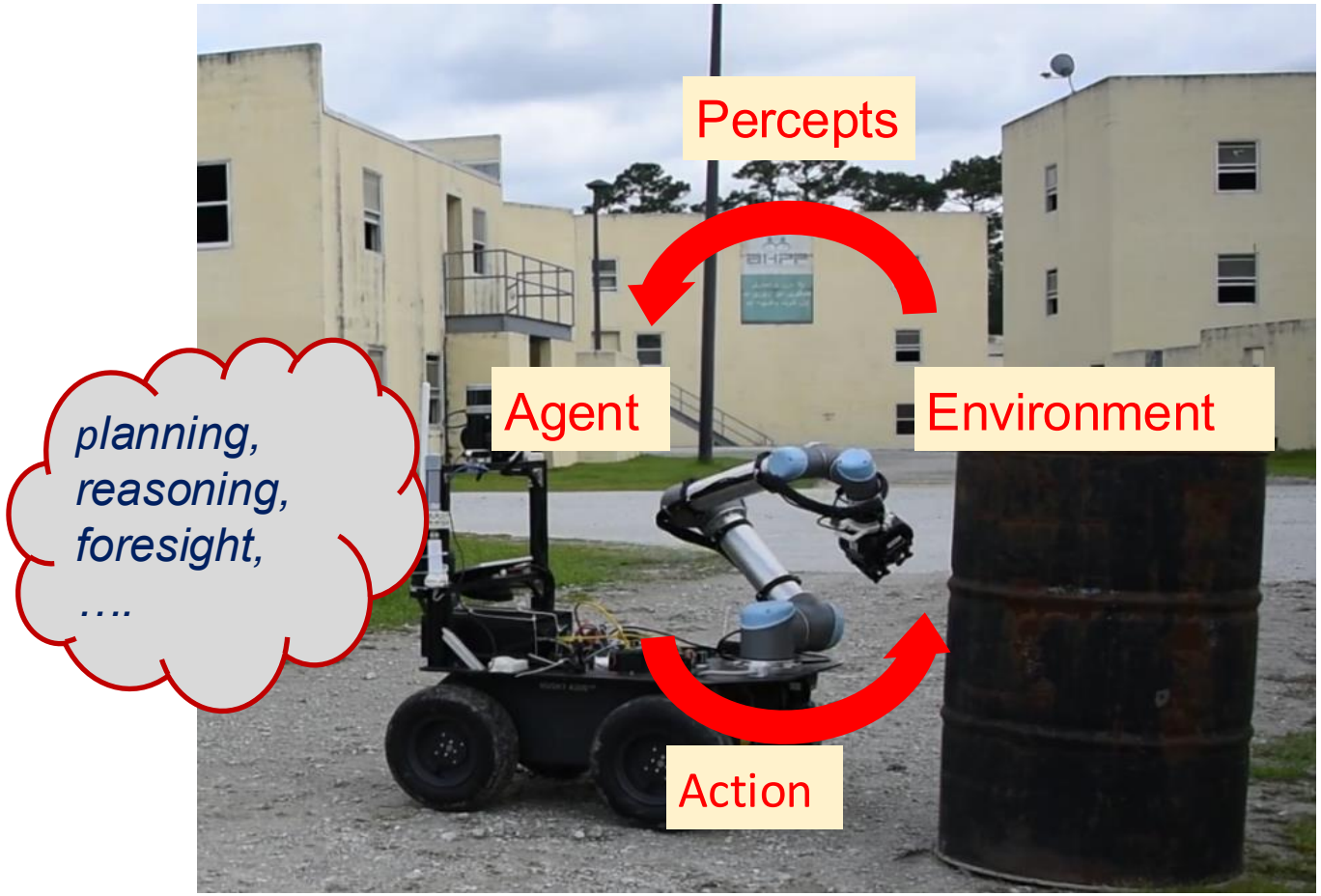


Monitoring power equipment
https://www.tagup.io/

Traffic Alert and Collision Avoidance Systems

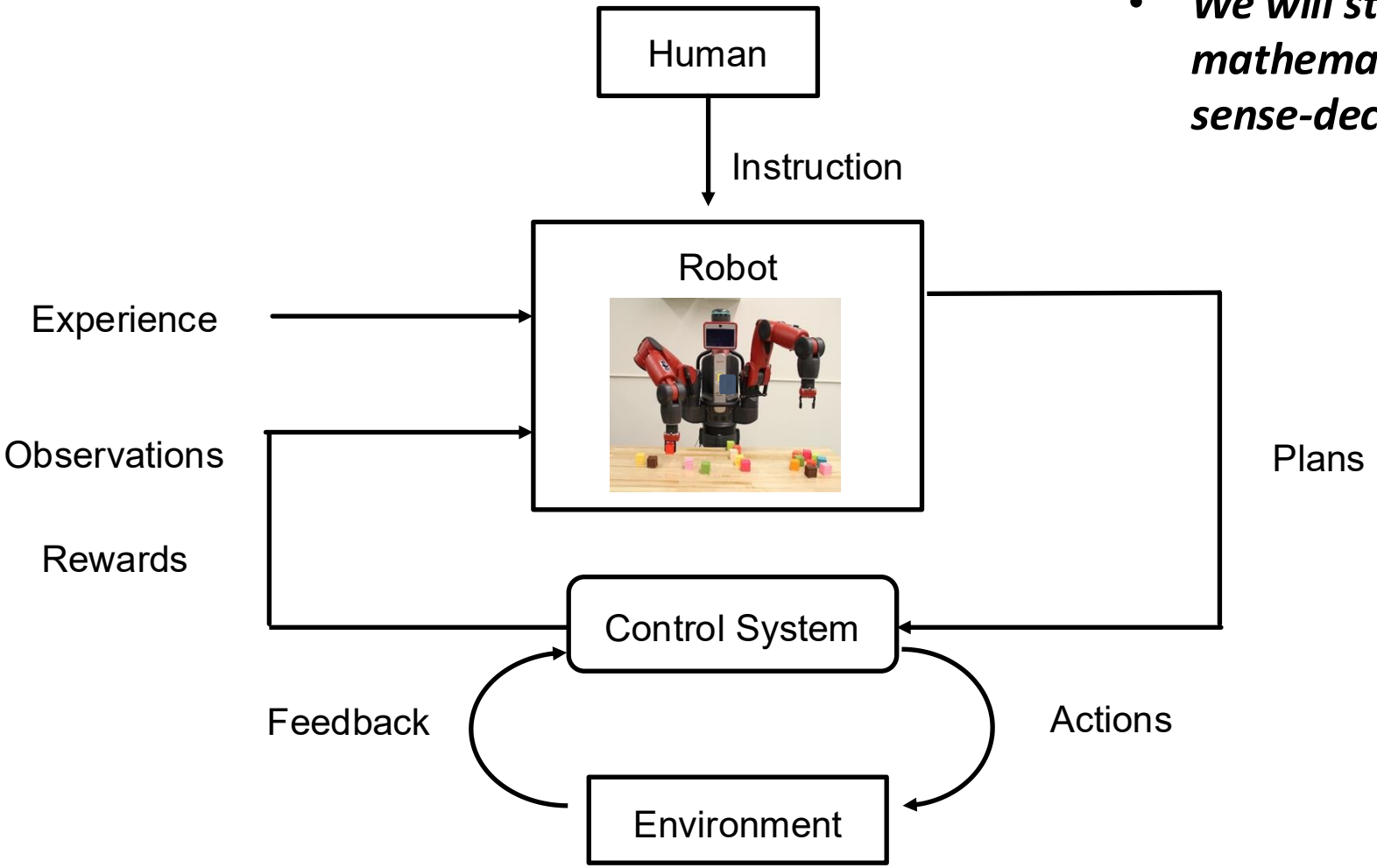
Agent Representation

- Observations
  - Sensor taks observations from the environment.
- Goal
  - Need for goal directed actions.
- Actions
  - Behavioral abstraction for actions. Pick, place, push …
- Objective: overall maximize expected utility over time.



Agent Representation

- ***Needs many aspects of AI and Learning to come together!***
- ***We will study the core mathematical formalisms for sense-decide-act cycle.***



An estimation task:

$$p(\text{plans} \mid \text{observations, instruction, experience})$$

Let’s take an example: Intelligent Manipulation

Pick			
Place			

Videos from research work in lab by : Sandeep Zachariah, Moksh Malhotra, Garima, Mh. Nadir. Collaborator: Prof. Rao. Supported by: DIA-COE.

Manipulation with Planning

Videos from research work in lab by : Sandeep Zachariah, Moksh Malhotra, Garima, Mh. Nadir. Collaborator: Prof. Rao. Supported by: DIA-COE.

Planning a Sequence of Actions

Generates the sequence of actions to take the robot from the initial state to the desired goal state.

$$\pi = [a_o, a_1, a_2, \ldots, a_{n-1}]$$

$$s_0 \xrightarrow{a_0} s_1 \xrightarrow{a_1} s_2 \xrightarrow{a_2} \ldots \xrightarrow{a_{n-2}} s_{n-1} \xrightarrow{a_{n-1}} s_n$$

$$\tau(s_i, a_i) \rightarrow s_{i+1}$$

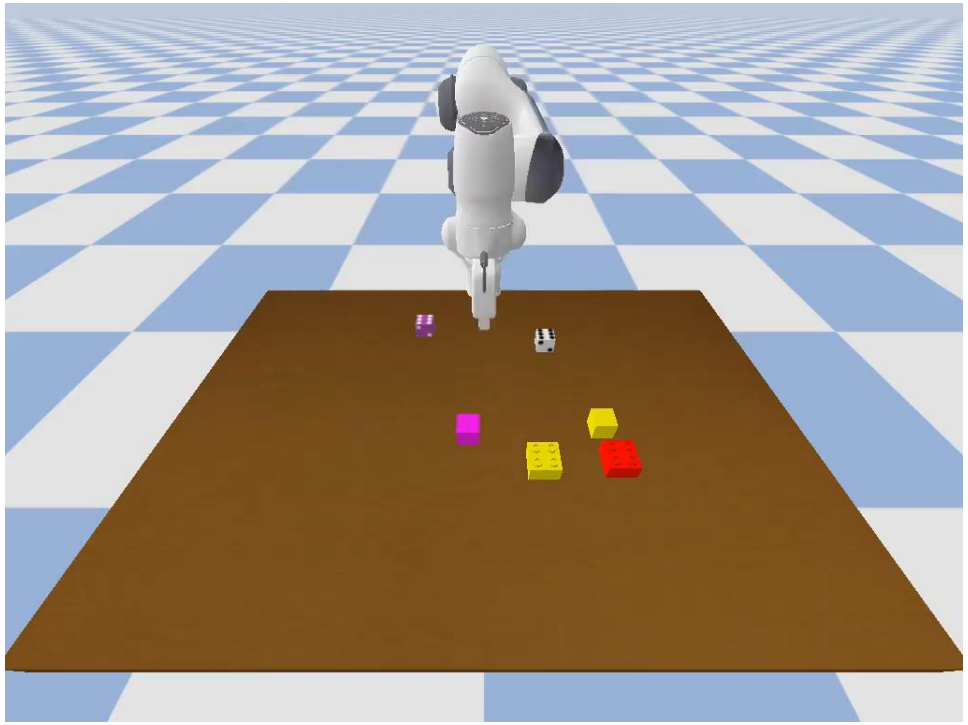
$$g_t \subseteq s_n$$

τ: transition function
π: plan
a_i: action at timestep i
s_i: state at timestep i
g_t: goal state

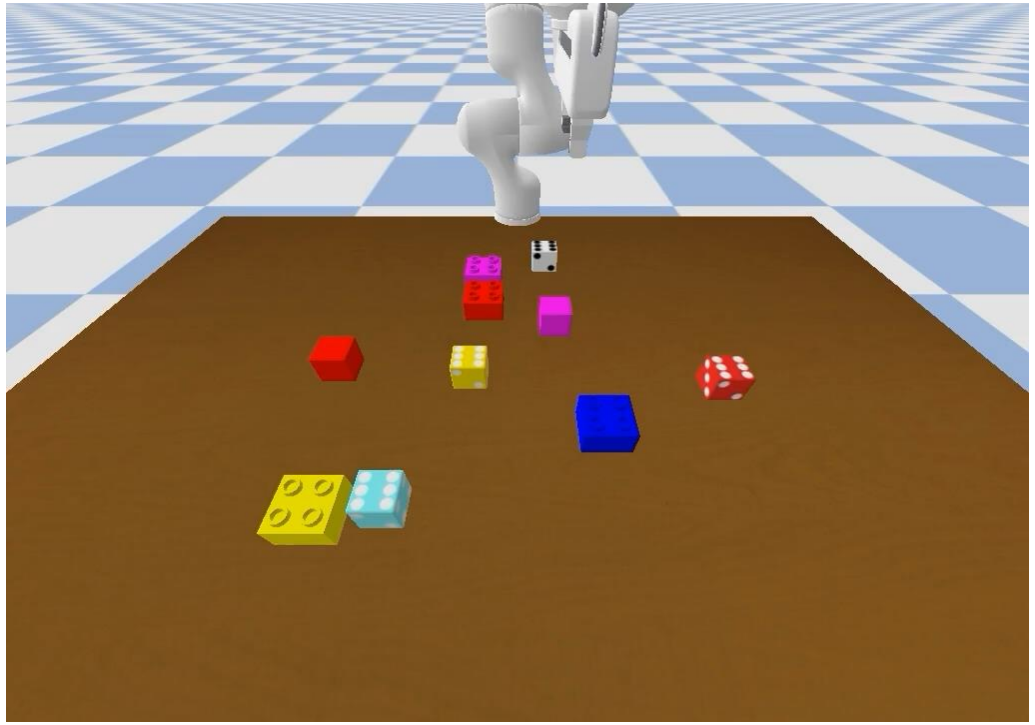
```
(:action pick
        :parameters (?obj - item ?loc - region)
        :precondition (and (gripperFree) (at ?obj ?loc)
                           (clear ?obj)
                           (onGround
?obj)
                           (isGraspable
?obj))
        :effect(and (not (at ?obj ?loc))
                    (not (gripperFree))
                    (not (onGround ?obj))
                    (holding ?obj)
(:action place          )
        :parameters (?obj - item ?loc - region)
        :precondition (holding ?obj)
        :effect(and (at ?obj ?loc)
                    (gripperFree)
                    (not (holding ?obj))
                    (onGround ?obj)
                )
        )
```

Reasoning and Planning

"Put the white dice above the yellow lego object and move the yellow cube on top of the white dice"



"Place the magenta cube to the left of red cube"
Sensor errors happen all the time!

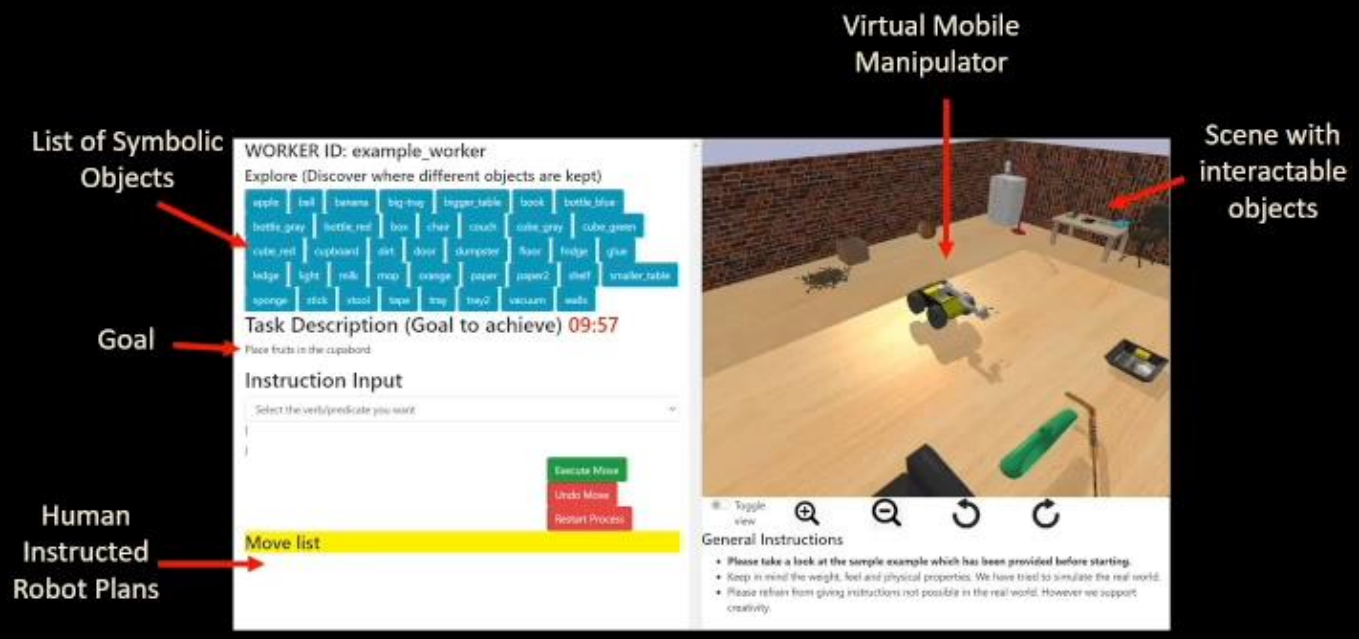


Core Idea: how to "teach" concepts to a robot (show demonstrations and let the robot imitate)

Videos from research work in lab by : Namasivayam K, Himanshu Gaurav Singh, Vishal Bindal, Arnav Tuli. Collaborator: Prof. Singla.

Performing Complex Tasks

Data Collection Interface

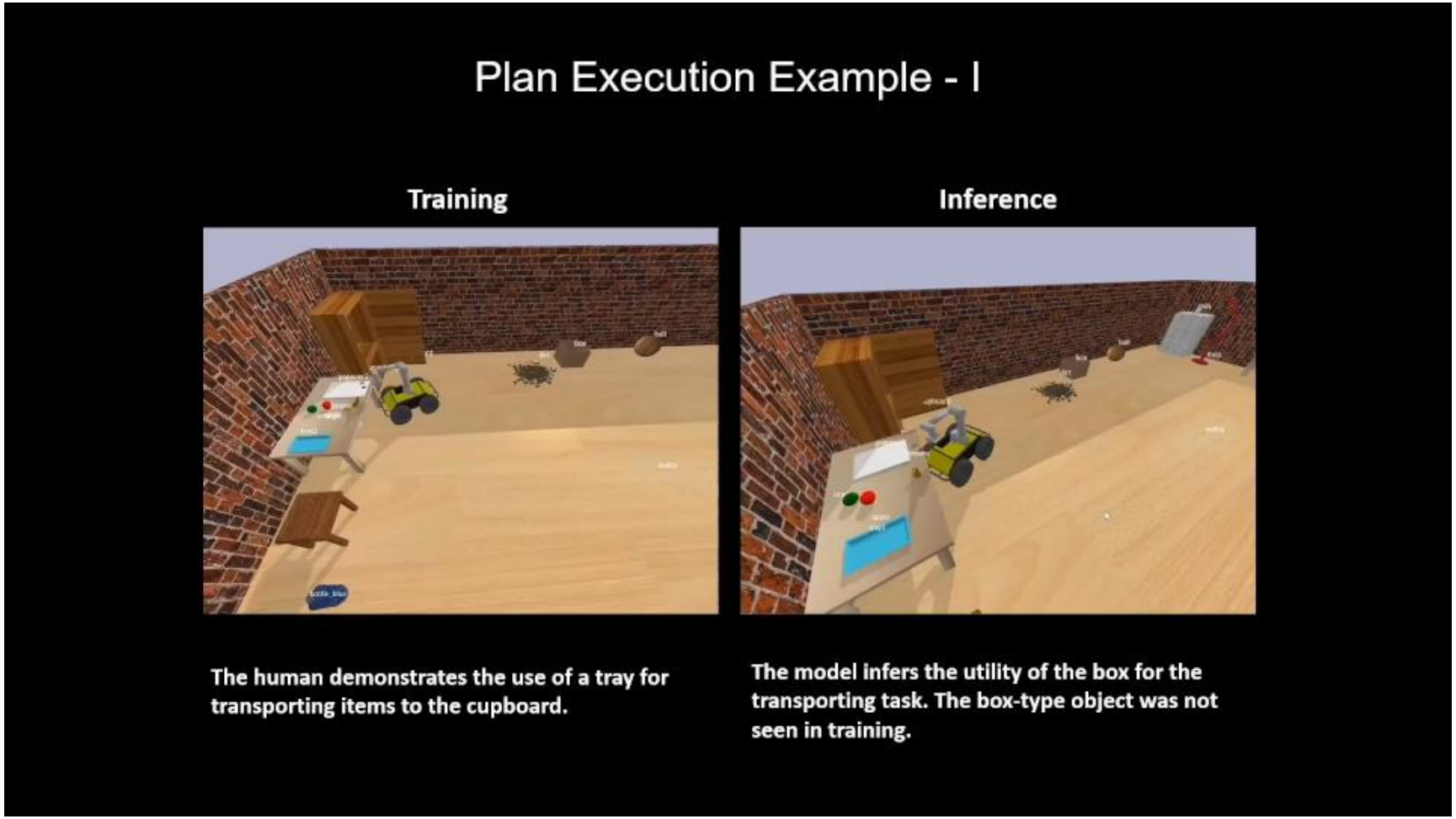


- A human teacher selects a goal (a high-level task) and specifies a symbolic plan for the robot.
- A coarse simulation is performed for each symbolic action in sequence.
- The world state, the goal and the instructed plan forms a datum for training.

Joint work with Shreshth Tuli, Rajas Bansal and Prof. Mausam.

Learning from human demonstration

The agent learns to imitate human plans. Also, generalize to new settings.



Learning from Demonstration

Environment, $\mathbf{S_t}$

Relational Knowledge Base (ConceptNet), **C**

HasA, RelatedTo, MadeOf() ….

Symbolic Goal, **G**

"Put the fruits in the fridge"

M(Θ)

Actions

Pick, place, drop, push, move

p(action | S_t, G, C)

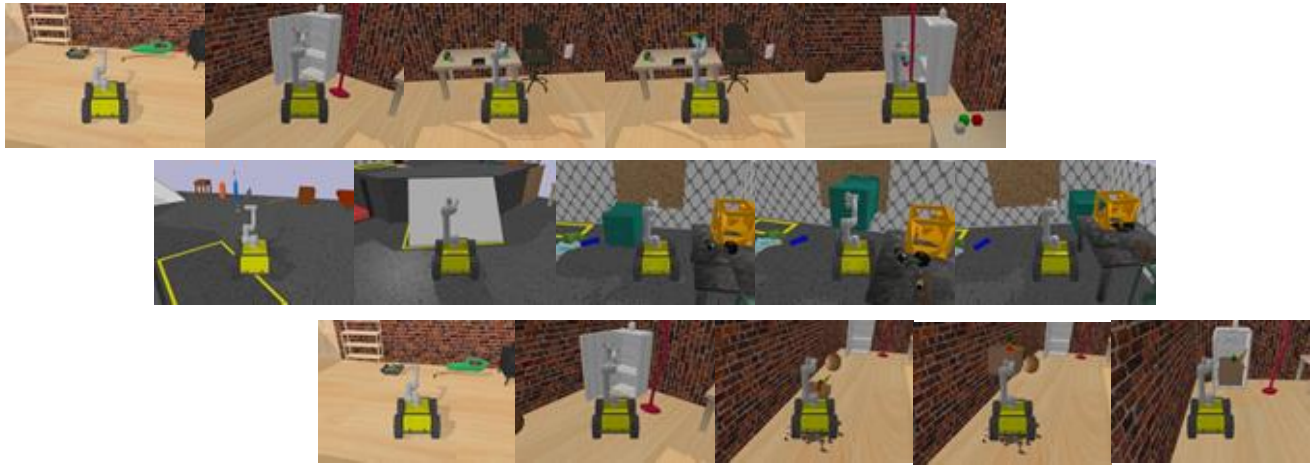
Model training: Human-instructed Robot Plans

$$< s_0^0, a_0^0, s_1^0, a_1^0, \ldots, s_g^0 >$$

$$< s_0^1, a_0^1, s_1^1, a_1^1, \ldots, s_g^1 >$$

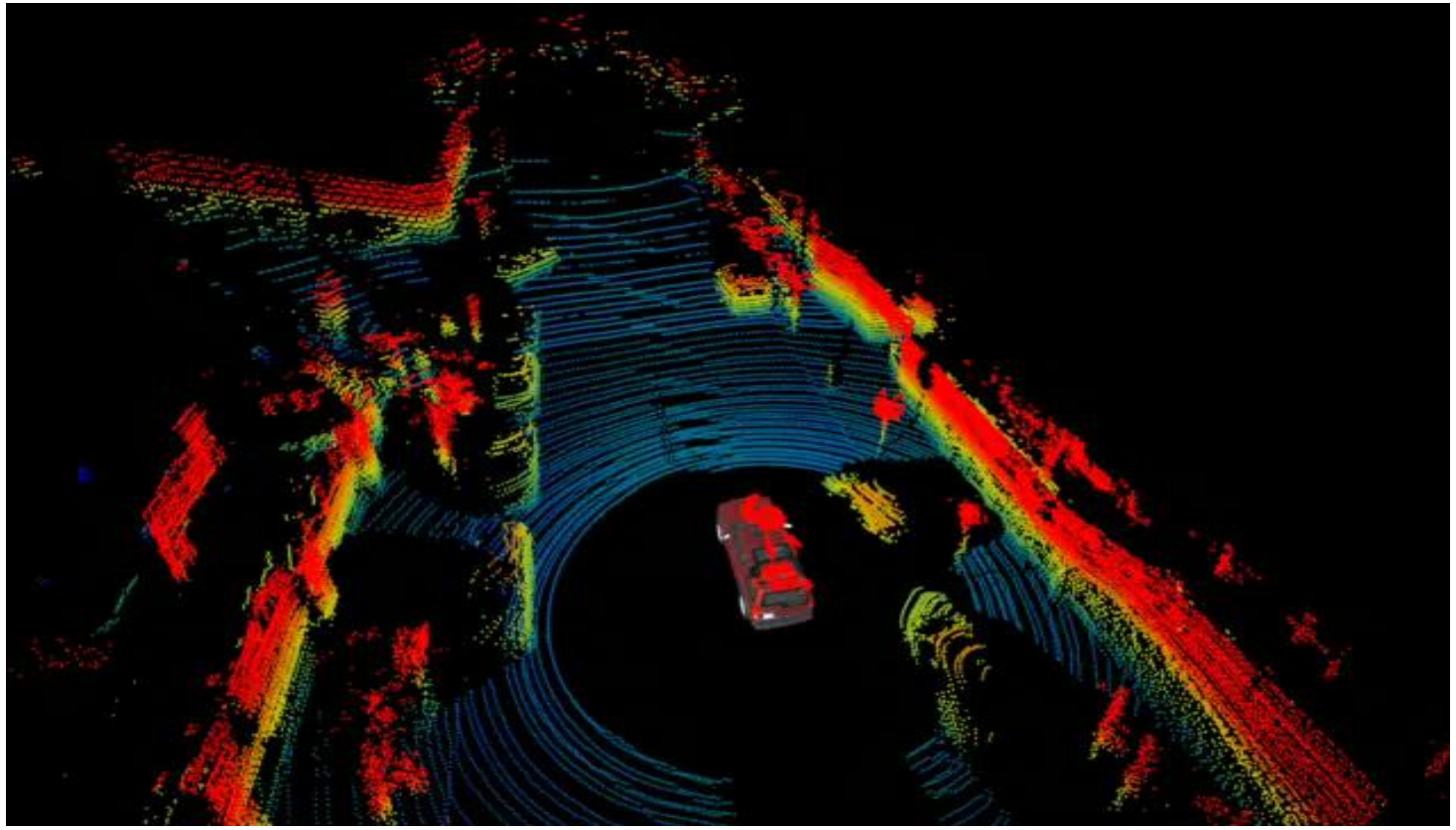
$$\vdots \qquad \vdots \qquad \vdots$$

$$< s_0^n, a_0^n, s_1^n, a_1^n, \ldots, s_g^n >$$



Challenge I: Observations are Noisy

- The agent observes the environment via sensors
- Noisy and uncertainty
  - Give a noisy and incomplete picture of the real world
- Inference task
  - Given observations estimated the true world state.



Challenge II: Partial-Observability

- Task Planning
  - In order to execute a task, we need to know some properties of the environment.
- Problem
  - Some of these properties are not directly observed.
  - Formally, some variables in the robot's state are not directly observed.
  - Example: the internal state of the barrel, inside of cupboards/boxes.
- Partial-observability
  - Formally, some variables in the robot's state are not directly observed.
  - Decision-making models must take this into account. Typically done by "exploration or information gathering actions"
  - Tools such as POMDP, information-theoretic planning

Intelligent Interaction with the World

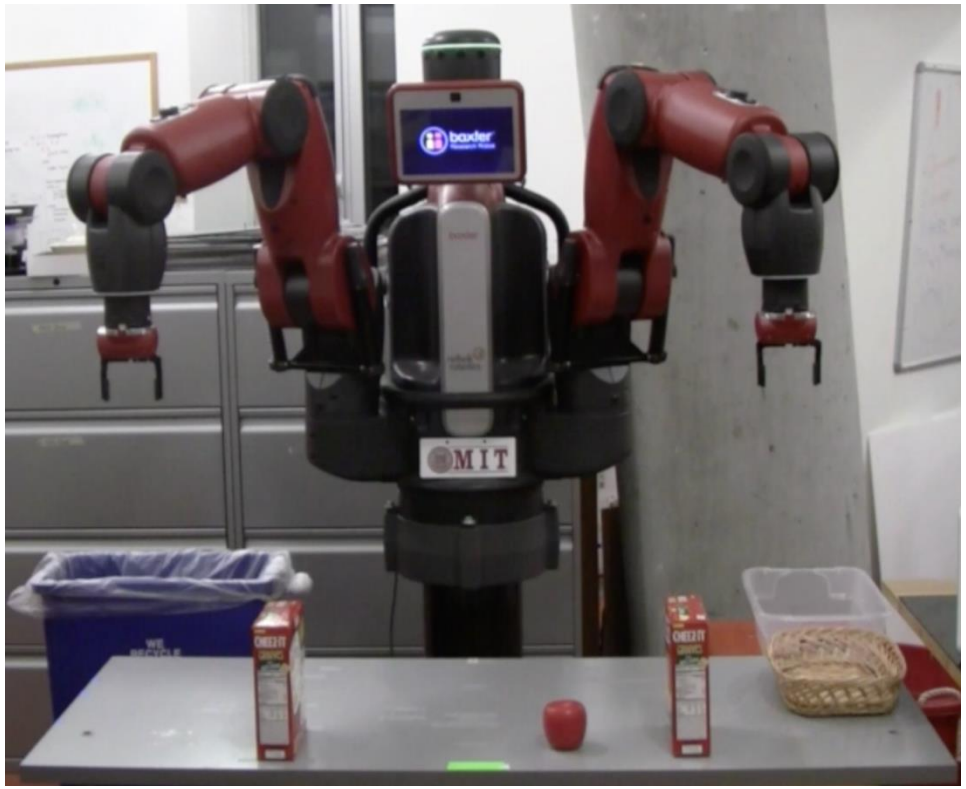
Experiment #2

A mobile manipulator is tasked with picking up a heavy case and is given correct information, incorrect information, and no information. The robot estimates the state of each object online until it converges to a solution. The natural language understanding model allows the robot to acquire the object given correct information more quickly but is also robust to incorrect information.

Active interaction with the environment to gather information for planning

Challenge III: Common sense knowledge

- We acquire a lot of knowledge from experience.
- AI agents often lack such knowledge.
- Examples:
  - To put an object inside the box, the lid must be removed.
  - Milk is an edible food item.
- You may have tried speaking to a Google home or automated reservation system and faced this issue.
  - Common problem in related dialogue systems also.



baxter
MIT

Challenge IV: Human Interaction

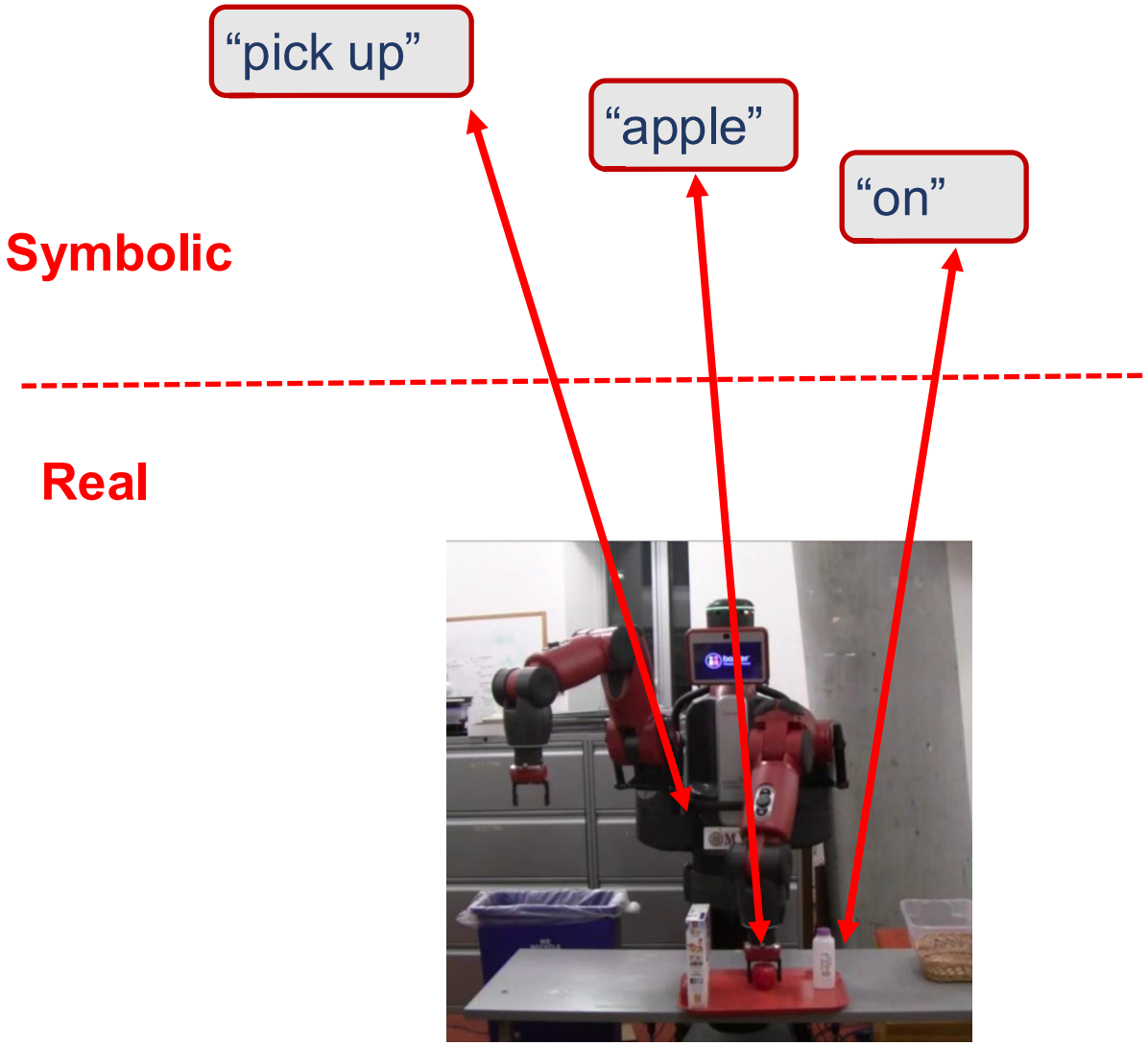
Embodied Agents to support the elderly

Computer controlled wheel chairs.

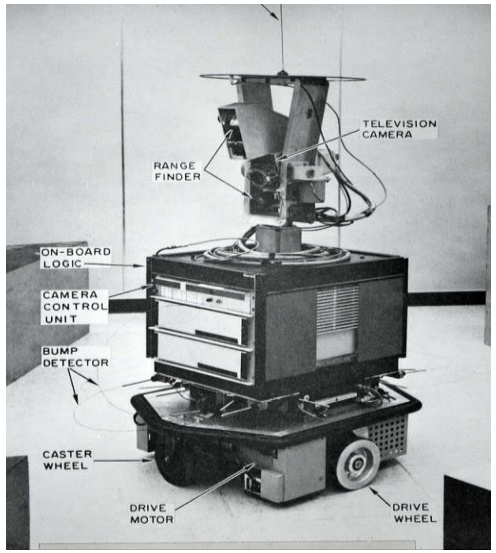
Intelligent agents to support persons with limitations

Challenge IV: Human Interaction

- Interpretable AI
  - Strong push towards AI system that are not black box.
  - Can explain their decision.
- Embodied AI context
  - The agent should be able to iteraction with a human.
- Computational problem
  - How can a program acquire the same semantic representation as a human.
  - Notions of objects, regions, interactions.
  - Also called “symbol grounding” problem



Several ideas in AI emerged from embodied systems



Visual recognition

Path planning

A* planner

Kalman filtering

State estimation

Localization /mapping,

Deep Learning for

perception

Inverse

reinforcement

Learning

Next Time

- This Class
  - Course Introduction
- Next Class
  - Agent Representation